BETSY — THE GAP ANALYSIS

How procurement works today, and how it changes with Betsy

WHAT THIS IS

This is the "before and after" story for Betsy. It shows how a small manufacturer's procurement runs today — all by hand, by one busy operations manager named Jenny — and how the same work looks once Betsy takes over the routine parts.

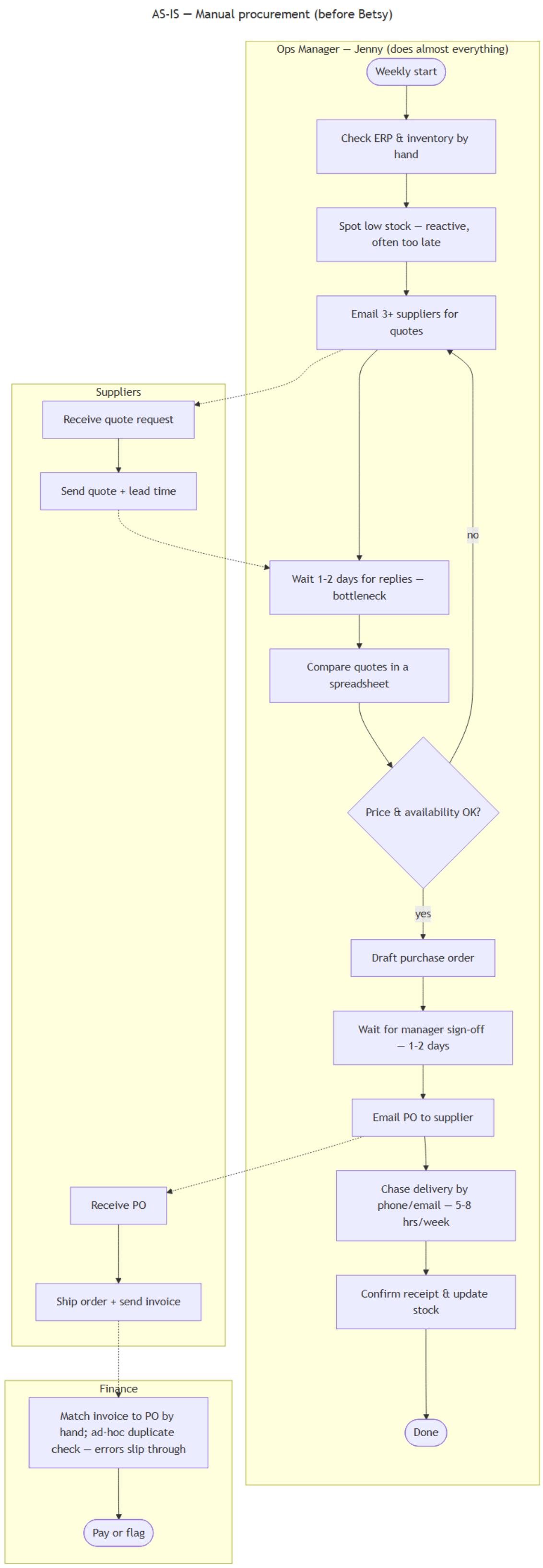
It is the starting point for the whole project: every design choice and every decision log points back to a problem shown here. A fully styled version lives in bpm_analysis.html; this document is the plain-text summary, with the diagrams pulled out as separate pictures so they can be reused anywhere.

THE BIG PICTURE

- Jenny spends about 30–35 hours a week on procurement firefighting today. With Betsy that drops to roughly 2–4 hours — only the decisions that truly need a person. That is about an 87% cut.
- Around 80% of the routine procurement work becomes automatic.
- Finding a supplier goes from 1–2 days of back-and-forth email to under 5 seconds, because all suppliers are checked at once.
- Every invoice is scanned for duplicates automatically — in the mock data this catches a repeated $1,470 invoice that would likely have been paid twice by hand.

BEFORE — HOW PROCUREMENT WORKS TODAY (AS-IS)

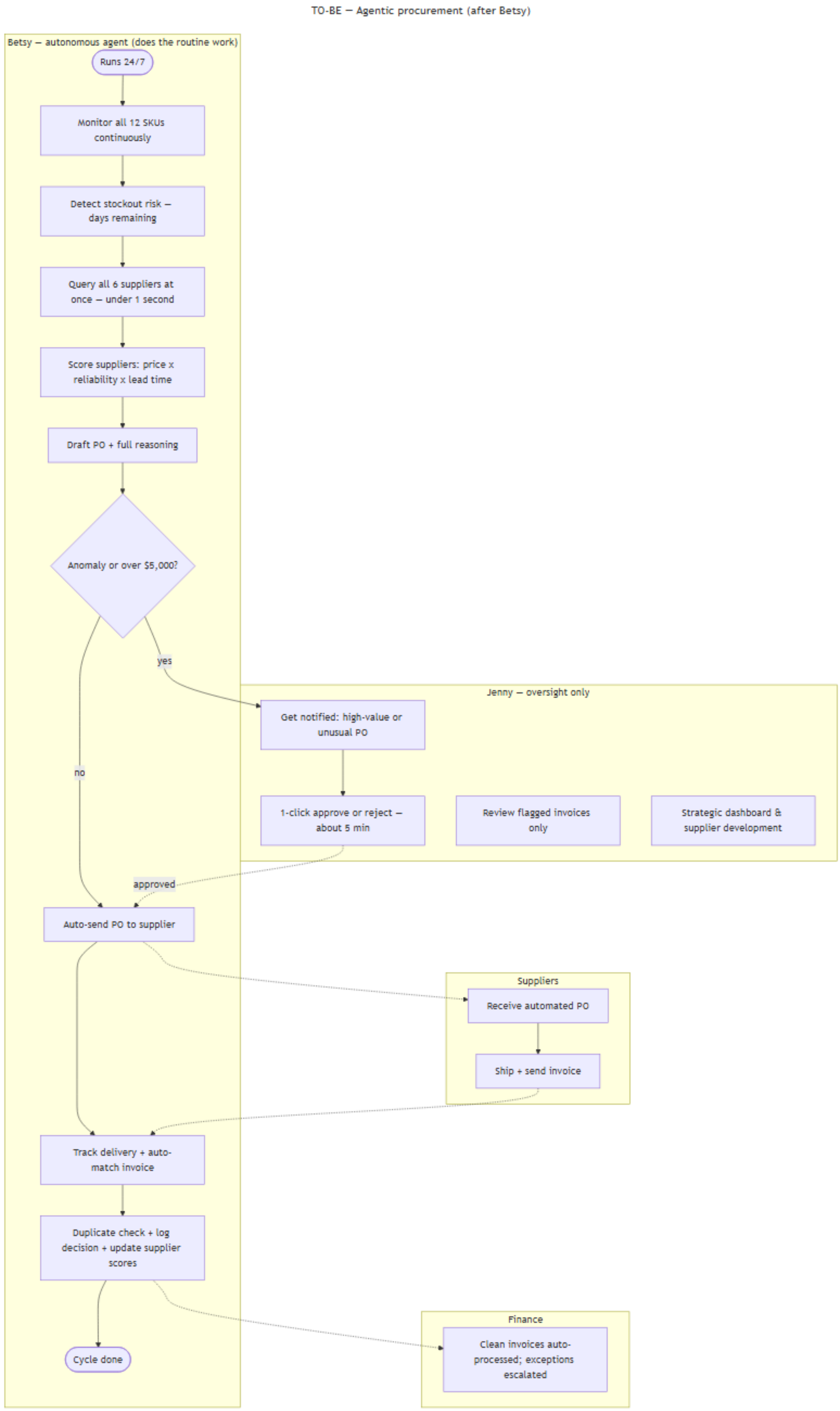
Jenny does almost everything herself, one step after another. She checks stock by hand, usually noticing a problem too late. She emails a few suppliers and waits a day or two for replies. She compares quotes in a spreadsheet, drafts the order, waits again for sign-off, emails the order out, then spends hours chasing the delivery. Finance matches invoices by hand, so duplicate bills can slip through. It is reactive, slow, and full of waiting.



See the picture: diagrams/gap-as-is.mmd

AFTER — HOW IT WORKS WITH BETSY (TO-BE)

Betsy watches stock around the clock and spots a shortage early. It checks all suppliers at once, scores them on price, reliability, and speed, and prepares an order with its reasoning written out. Routine orders go straight through; anything high-value or unusual is handed to Jenny for a quick yes/no. Deliveries and invoices are matched automatically, and duplicates are flagged before payment. Jenny is freed up for the strategic work she was actually hired to do.

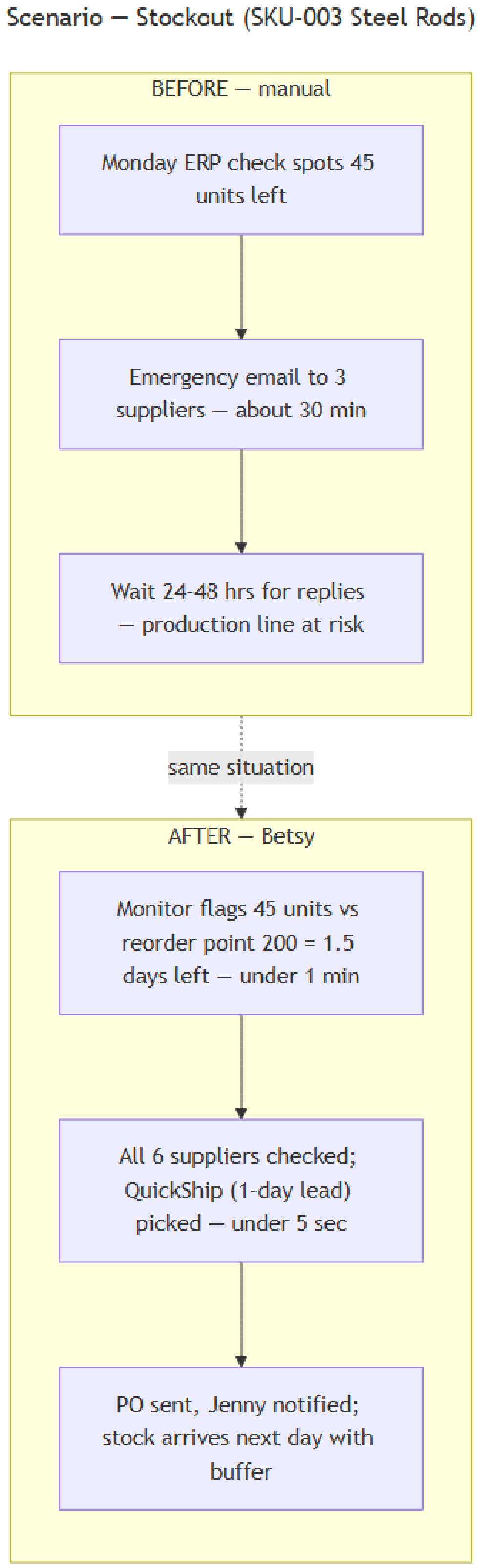


See the picture: diagrams/gap-to-be.mmd

THREE REAL EXAMPLES

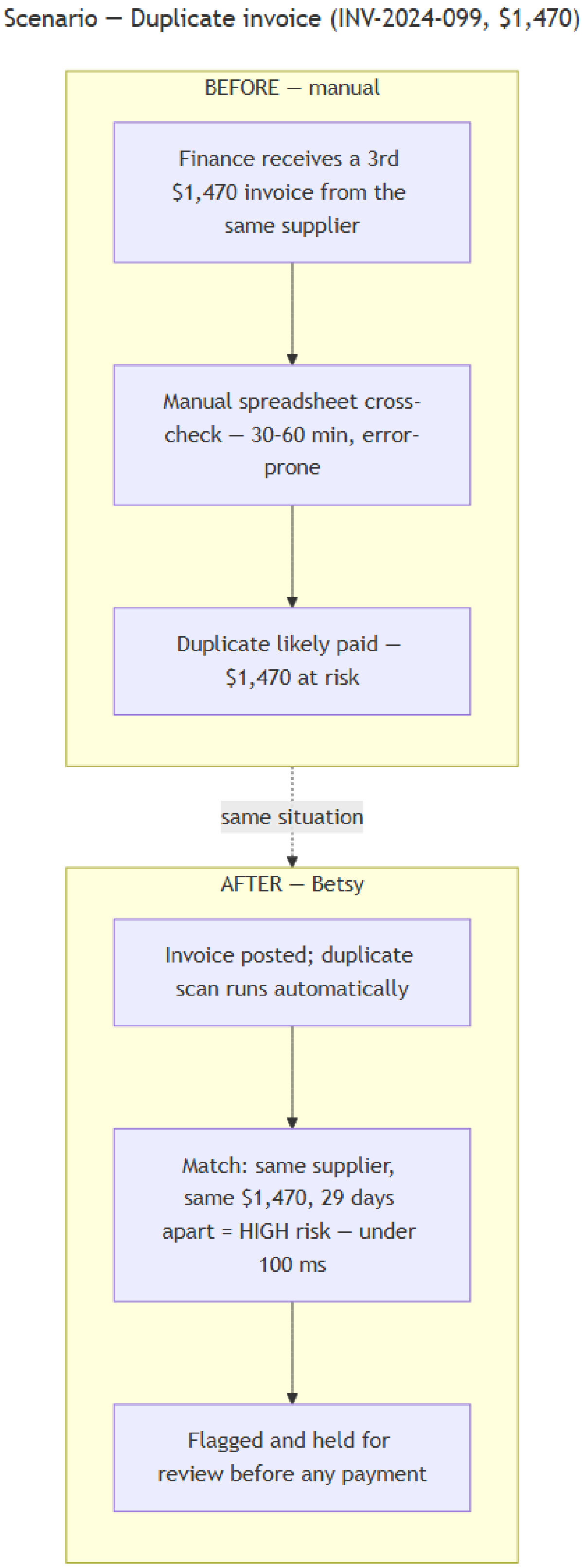
These are the three situations Betsy is tested against. Each shows the same event handled the old way and the Betsy way.

1. A near stockout on Steel Rods. By hand, it is spotted on a Monday with the line at risk and a scramble to email suppliers. With Betsy, it is caught within a minute and an order is on its way in seconds.



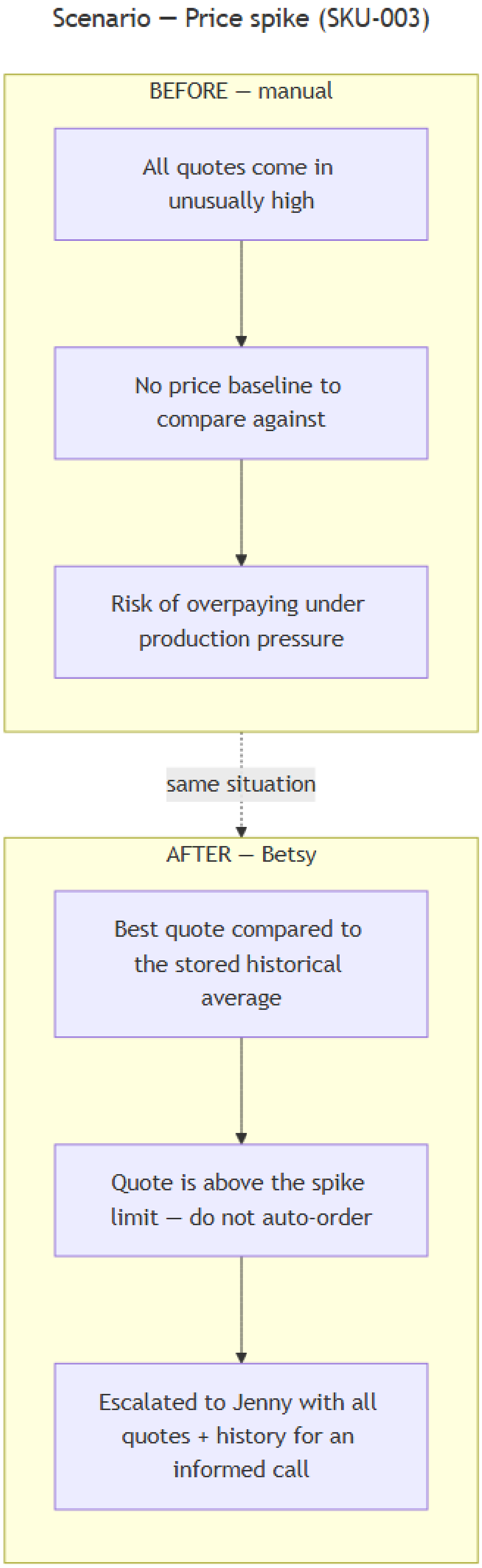
See the picture: diagrams/gap-scenario-stockout.mmd

2. A duplicate invoice. By hand, a third identical $1,470 bill is easy to miss and likely paid. With Betsy, it is flagged as a high risk in well under a second and held before any payment.



See the picture: diagrams/gap-scenario-duplicate-invoice.mmd

3. A price spike. By hand, there is no baseline, so there is a real risk of overpaying under pressure. With Betsy, the quote is compared to the historical average and escalated to Jenny with the full context.



See the picture: diagrams/gap-scenario-price-spike.mmd

TIME SAVED EACH WEEK

Before Betsy, a typical week is roughly: 4–5 hours checking stock, 6–8 hours on supplier emails, 2–3 hours comparing quotes, 4–5 hours drafting and chasing approvals, 5–8 hours chasing deliveries, 3–4 hours on invoices, and 4–6 hours of Monday-morning firefighting — about 30–39 hours.

After Betsy, almost all of that is automatic. What is left is roughly half an hour to an hour approving high-value orders, the same for reviewing flagged invoices, and one to two hours on strategy — about 2–4 hours. That is roughly 27–35 hours given back every week.

WHY THIS MATTERS (AND WHERE IT IS USED)

This before/after picture is the reason the rest of the project exists. The decision logs each point back to a specific gain shown here — the approval flow keeps Jenny's one human checkpoint, the duplicate detector closes the invoice-error gap, the supplier scoring replaces gut-feel sourcing, and so on. The diagrams above are meant to be dropped straight into those logs as the justification for each piece of work.

WHERE THIS LIVES

- bpm_analysis.html the full styled before/after report
- scenarios/ the three tested situations, as data
- mock_data/ the 12 SKUs, 6 suppliers, and invoices

DIAGRAMS FOR THIS DOCUMENT

- diagrams/gap-as-is.mmd the manual process (before)
- diagrams/gap-to-be.mmd the Betsy process (after)
- diagrams/gap-scenario-stockout.mmd near stockout, before/after
- diagrams/gap-scenario-duplicate-invoice.mmd duplicate bill, before/after
- diagrams/gap-scenario-price-spike.mmd price jump, before/after